Simultaneous analysis of serotonin, melatonin, piceid and resveratrol in fruits using liquid chromatography tandem mass spectrometry.

An analytical method was developed for the simultaneous quantification of serotonin, melatonin, trans- and cis-piceid, and trans- and cis-resveratrol using reversed-phase high performance liquid chromatography coupled to mass spectrometry (HPLC-MS) with electrospray ionization (ESI) in both positive and negative ionization modes. HPLC optimal analytical separation was achieved using a mixture of acetonitrile and water with 0.1% formic acid as the mobile phase in linear gradient elution. The mass spectrometry parameters were optimized for reliable quantification and the enhanced selectivity and sensitivity selected reaction monitoring mode (SRM) was applied. For extraction, the direct analysis of initial methanol extracts was compared with further ethyl acetate extraction. In order to demonstrate the applicability of this analytical method, serotonin, melatonin, trans- and cis-piceid, and trans- and cis-resveratrol from 24 kinds of commonly consumed fruits were quantified. The highest serotonin content was found in plantain, while orange bell peppers had the highest melatonin content. Grape samples possessed higher trans- and cis-piceid, and trans- and cis-resveratrol contents than the other fruits. The results indicate that the combination of HPLC-MS detection and simple sample preparation allows the rapid and accurate quantification of serotonin, melatonin, trans- and cis-piceid, and trans- and cis-resveratrol in fruits.